

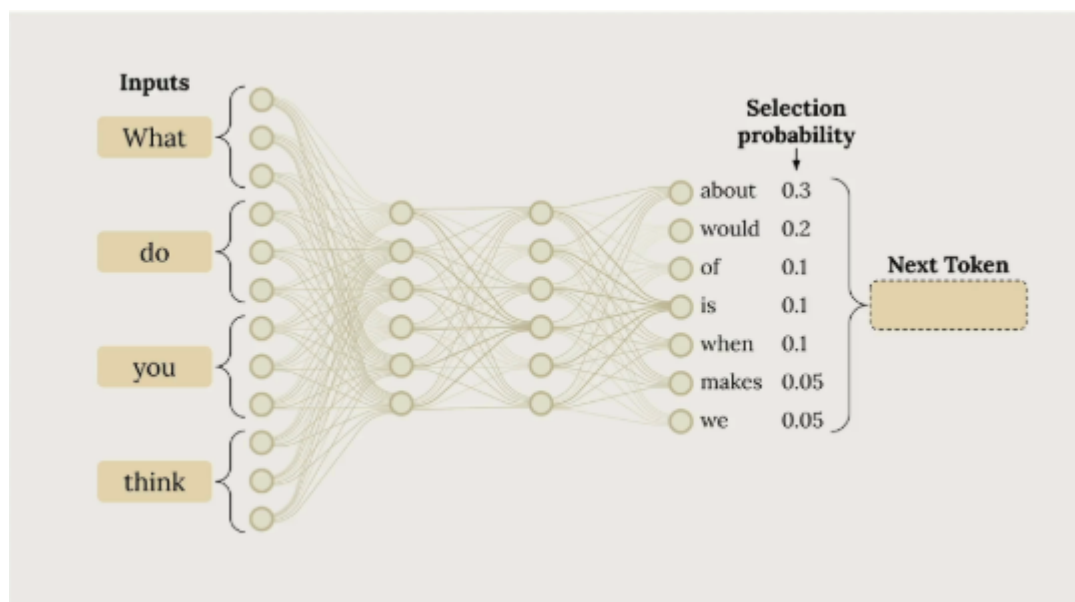
Temperature

Temperature is a powerful parameter that controls how predictable or creative Claude's responses will be. Understanding how to use it effectively can dramatically improve your AI applications.

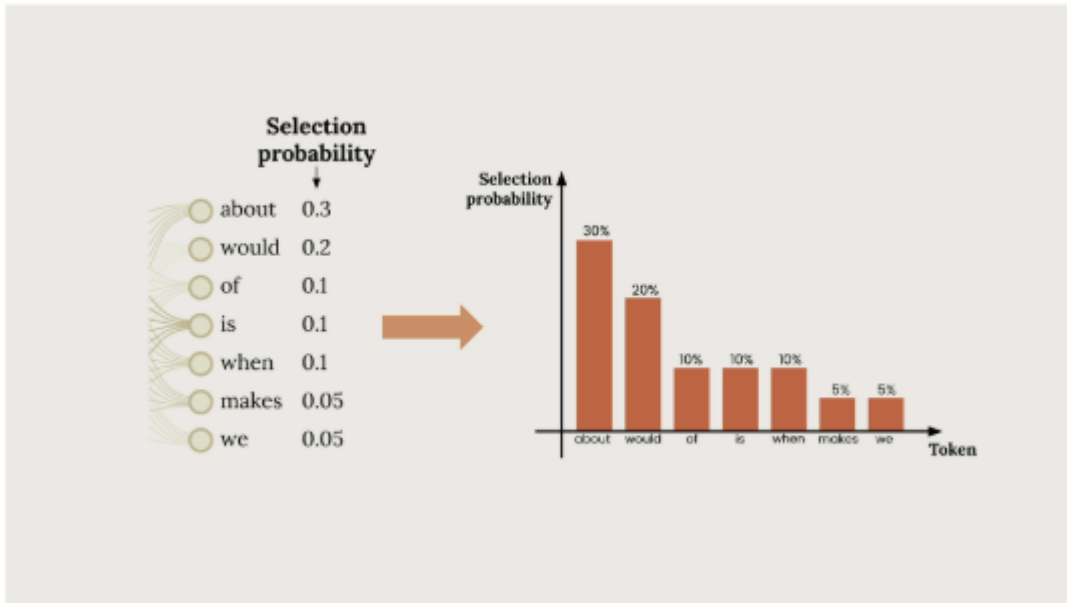
How Claude Generates Text

Before diving into temperature, it helps to understand Claude's text generation process. When you send Claude a prompt like "What do you think?", it goes through three key steps:

- Tokenization - Breaking your input into smaller chunks
- Prediction - Calculating probabilities for possible next words
- Sampling - Choosing a token based on those probabilities

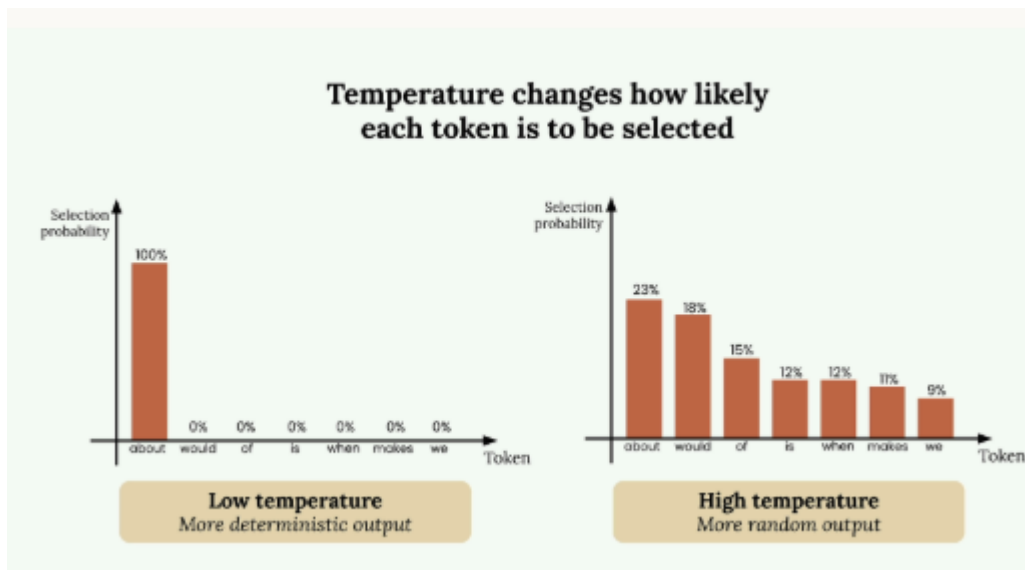


In this example, Claude might assign a 30% probability to "about", 20% to "would", 10% to "of", and so on. The model then selects one token and repeats this entire process to build complete sentences.



What Temperature Does

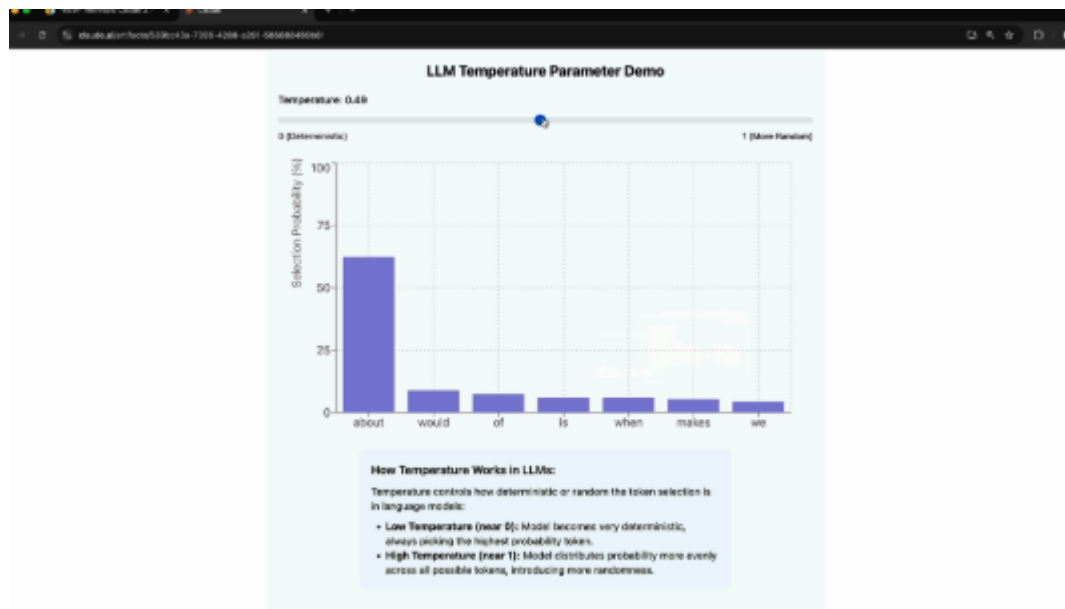
Temperature is a decimal value between 0 and 1 that directly influences these selection probabilities. It's like adjusting the "creativity dial" on Claude's responses.



At low temperatures (near 0), Claude becomes very deterministic - it almost always picks the highest probability token. At high temperatures (near 1), Claude distributes probability more evenly across options, leading to more varied and creative outputs.

Interactive Temperature Demo

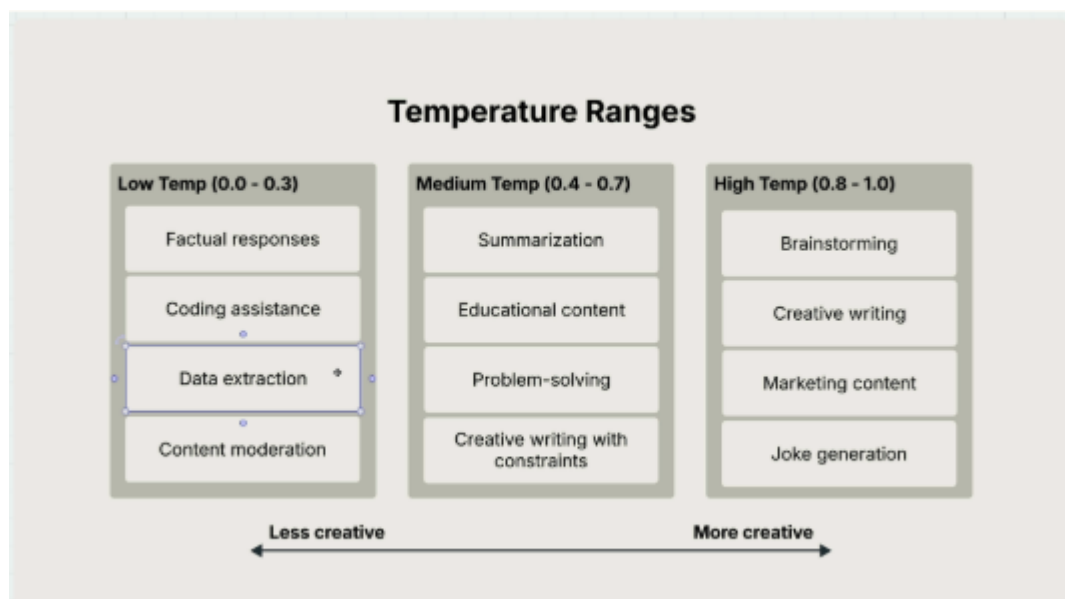
You can see temperature in action with Claude's interactive demo. Watch how the probability distribution changes as you adjust the temperature slider:



At temperature 0.0, "about" gets 100% probability - completely deterministic. At temperature 1.0, probabilities spread more evenly across all possible tokens, introducing randomness and creativity.

Choosing the Right Temperature

Different tasks call for different temperature ranges:



Low Temperature (0.0 - 0.3)

- Factual responses
- Coding assistance
- Data extraction
- Content moderation

Medium Temperature (0.4 - 0.7)

- Summarization
- Educational content
- Problem-solving
- Creative writing with constraints

High Temperature (0.8 - 1.0)

- Brainstorming
- Creative writing
- Marketing content
- Joke generation

Implementing Temperature in Code

Adding temperature support to your chat function is straightforward. Here's how to modify your existing function:

```
def chat(messages, system=None, temperature=1.0):
    params = {
        "model": model,
        "max_tokens": 1000,
        "messages": messages,
        "temperature": temperature
    }

    if system:
        params["system"] = system

    message = client.messages.create(**params)
    return message.content[0].text
```

The key changes are adding `temperature=1.0` as a parameter and including `"temperature": temperature` in the params dictionary.

Testing Temperature Effects

To see temperature in action, try generating movie ideas with different settings:

```
# Low temperature - more predictable
answer = chat(messages, temperature=0.0)
```

```
# High temperature - more creative  
answer = chat(messages, temperature=1.0)
```

At temperature 0.0, you might consistently get responses like "A time-traveling archaeologist must prevent ancient artifacts from being stolen." At temperature 1.0, you'll see much more variety in themes, characters, and plot elements.

Key Takeaways

Remember that temperature doesn't guarantee different outputs - it just changes the probability of getting them. Even at high temperatures, Claude might occasionally produce similar responses. The key is matching your temperature choice to your specific use case:

- Need consistent, factual responses? Use low temperature
- Want creative brainstorming? Dial up the temperature
- Somewhere in between? Medium temperatures work well for most general tasks

Temperature is one of the most practical parameters you can adjust to fine-tune Claude's behavior for your specific needs.